

amrsamir.me — Project Architecture & Site Map

This document serves as a comprehensive brief and map of the `amrsamir.me` project. It is intended for software engineers, stakeholders, and developers who need to understand the structural, architectural, and technological foundation of the platform.

1. System Architecture & Tech Stack

The platform is designed as a modern, serverless, and highly performant web application.

Frontend

- **Framework:** `Vite` (Vanilla JavaScript + HTML/CSS). Chosen for lightning-fast builds and zero-overhead performance.
- **Styling:** Vanilla CSS utilizing modern variables, CSS grid/flexbox, and micro-animations (like glassmorphism and abstract floating elements).

Backend & Database (BaaS)

- **Provider:** `Supabase` (PostgreSQL).
- **Authentication:** Supabase Auth (Email/Password) used to secure the Admin Dashboard.
- **Data Access:** Row Level Security (RLS) is strictly enforced. Public users can read settings and submit forms, but only authenticated users can edit settings or view the inbox.

Analytics & Tracking

- **PostHog:** Integrated via `posthog-js` for comprehensive product analytics, session recordings, pageviews, and heatmaps.
- **Vercel Analytics:** Built-in Vercel analytics package (`@vercel/analytics`) for basic traffic and web vitals monitoring.

Hosting & CI/CD

- **Provider:** `Vercel`.
 - **Deployment Flow:** Any push to the main GitHub branch triggers an automatic production build and deployment to Vercel.
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2. Site Map & Routing

The application is structured as a Multi-Page Application (MPA) using Vite's Rollup configuration.

Public Facing Pages

- `/` (**Home Hub**)
`index.html` — The main entry point. Features the dynamic background, the command palette (`CMD+K`), and a grid leading to the different personas. Also includes the dynamically fetched Substack newsletter embed.
- `/events` (**Events Persona**)
`events/index.html` — Portfolio tailored for events management and production. Contains a contact form that pushes data to Supabase.
- `/marketing` (**Marketing Persona**)
`marketing/index.html` — Portfolio tailored for brand strategy and digital marketing. Contains a

contact form that pushes data to Supabase.

- **/ai (AI & Tech Persona)**
`ai/index.html` — Portfolio tailored for tech innovations and AI agents. Contains a contact form that pushes data to Supabase.
- **404.html**
Custom 'Not Found' page for handling broken routes.

Secure Internal Pages

- **/admin (Admin Control Panel)**
`admin/index.html` — Secure dashboard exclusively for the owner.
 - **Login Screen:** Prevents unauthorized access.
 - **Settings View:** Allows the owner to dynamically update global links (WhatsApp, Email, LinkedIn, Substack). Updates here are instantly reflected across all public pages.
 - **Inbox View:** A real-time feed of all contact form submissions sent from the public persona pages.
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3. Database Schema

The PostgreSQL database (managed via Supabase) contains two primary tables:

`public.settings`

Stores the global dynamic links used across the platform.

- `id` (bigint, PK)
- `whatsapp` (text)
- `email` (text)
- `linkedin` (text)
- `substack` (text)
- `updated_at` (timestamp)

RLS Rules: Public can read (SELECT). Only Auth can UPDATE.

`public.form_submissions`

Stores messages sent by users from the website.

- `id` (bigint, PK)
- `name` (text)
- `email` (text)
- `message` (text)
- `created_at` (timestamp)

RLS Rules: Public can insert (INSERT). Only Auth can read (SELECT) or DELETE.

4. Environment Variables

To run this project, the following variables must be present in the local `.env` file and the Vercel Production Environment Settings:

```
# Required to connect the frontend to the database
VITE_SUPABASE_URL=https://<your-project-id>.supabase.co
VITE_SUPABASE_ANON_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXLTJ0eSIsImF1dG8iOiJkZWZhdGVkaW9udC5jb20iLCJpcyI6ImNpdGUiLCJhdWQiOiJkZWZhdGVkaW9udC5jb20iLCJyb2l0IjoiYWRtaWUiLCJleSI6MTYwMDE1OTQxMCwiZXhwIjoxNjA0MDg3ODExfQ..

# Note: PostHog API key is public and hardcoded in main.js
```